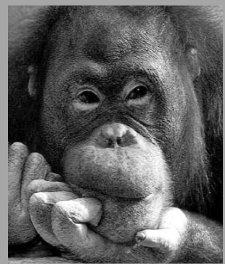
   

Phenotype = Genetics + Environment + GxE

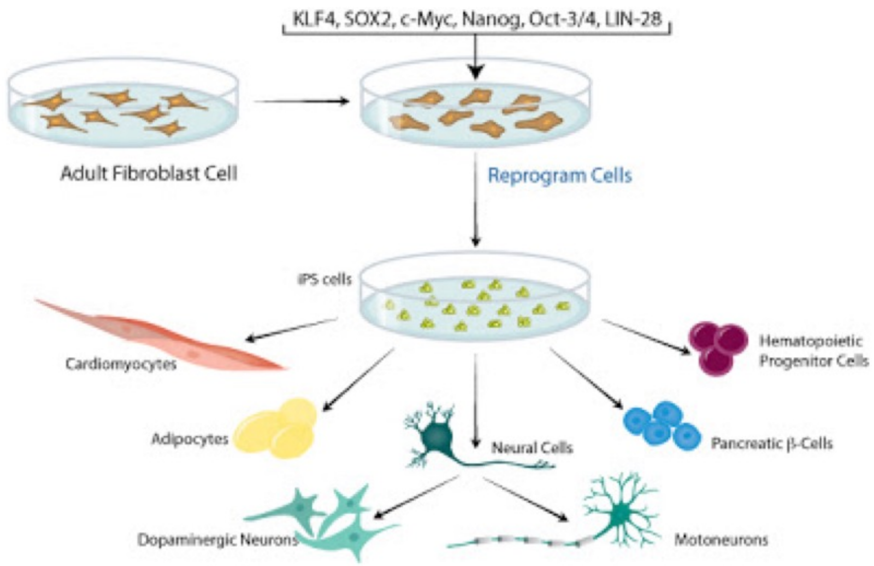
$(V_p = V_g + V_e + V_{gxe})$

"If, as is generally implied in genetic work (although not often explicitly stated), all of the genes are active all the time; and if the characters of the individual are determined by the genes, then why are not all the cells of the body exactly alike? "

T. H. Morgan, Nobel lecture, 1934

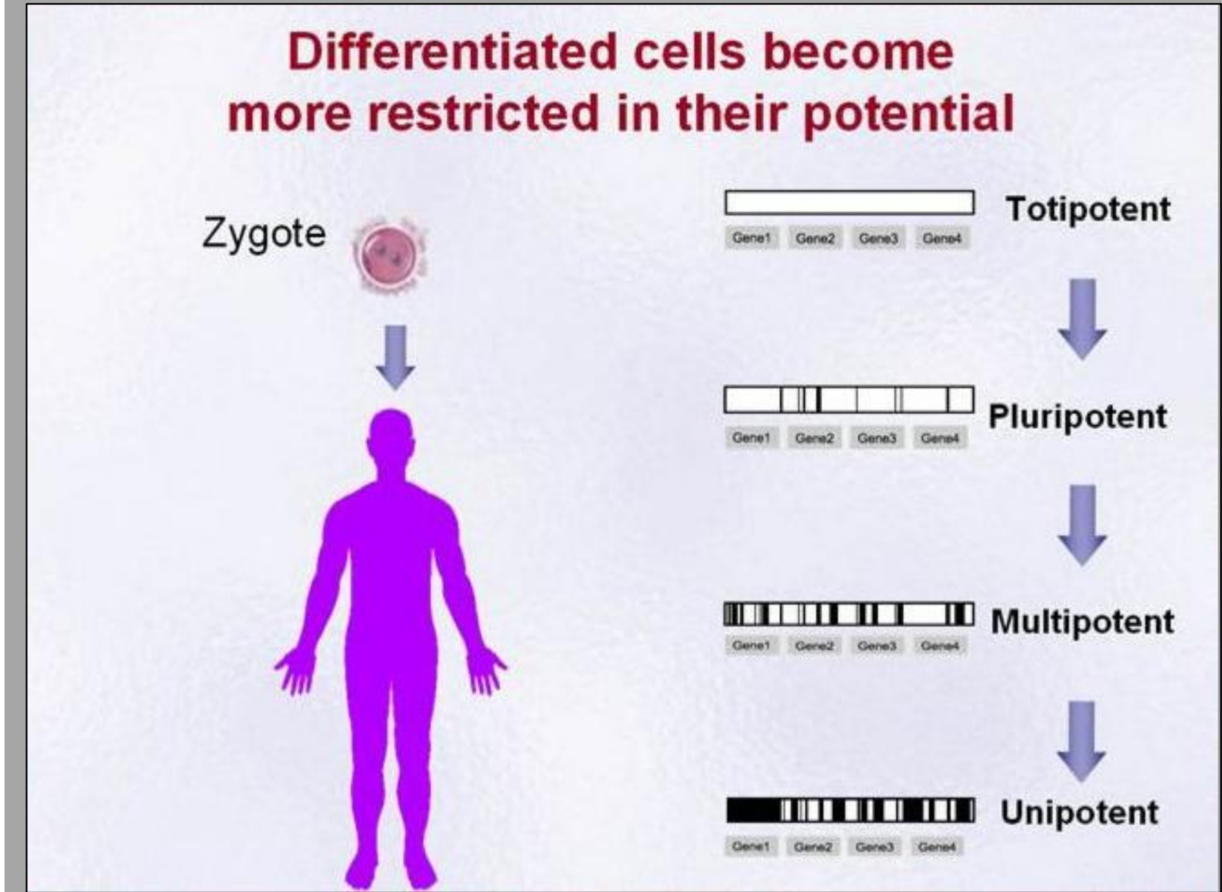
Células diferenciadas podem voltar a ser multipotentes

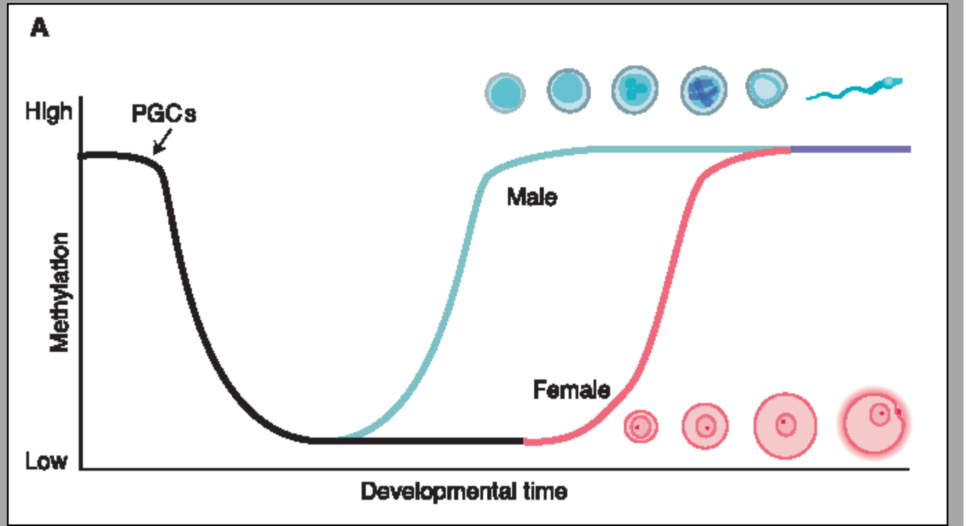
Briggs and King, 1952
Gurdon, 1960s



Yamanaka, 2006

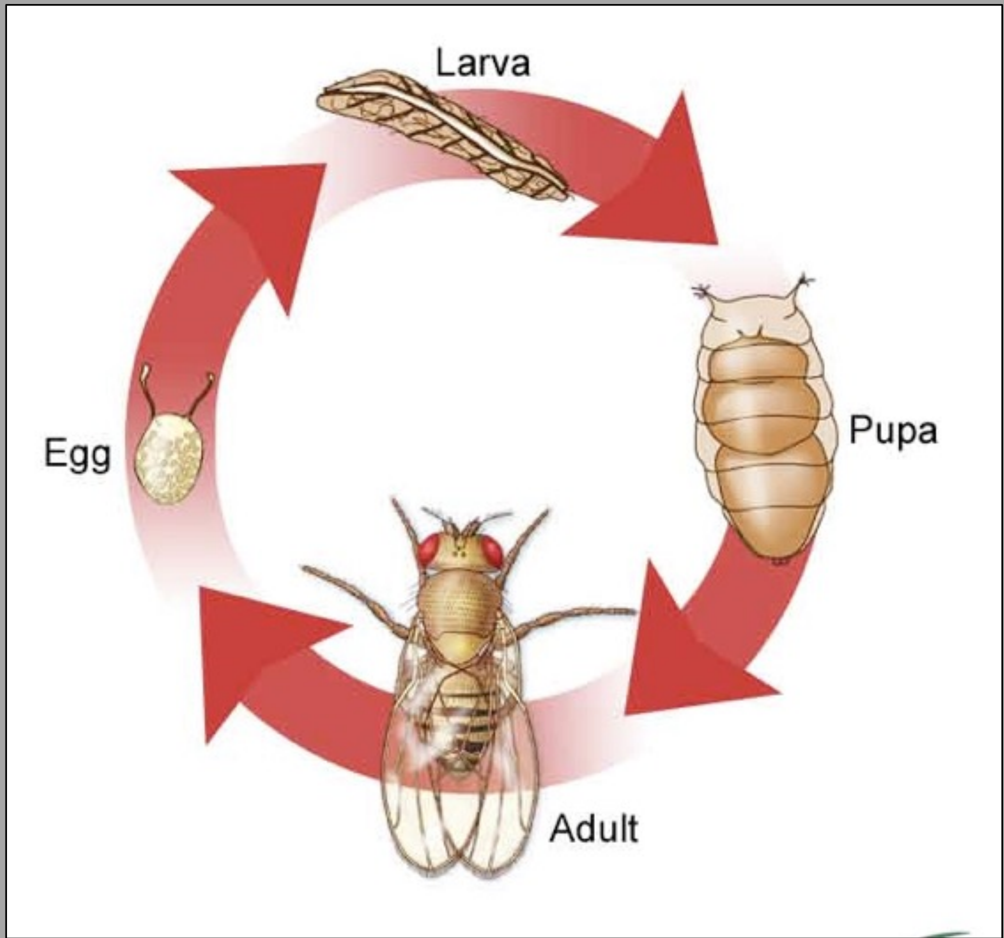
DNA metilado da linha germinal zigoto ao adulto





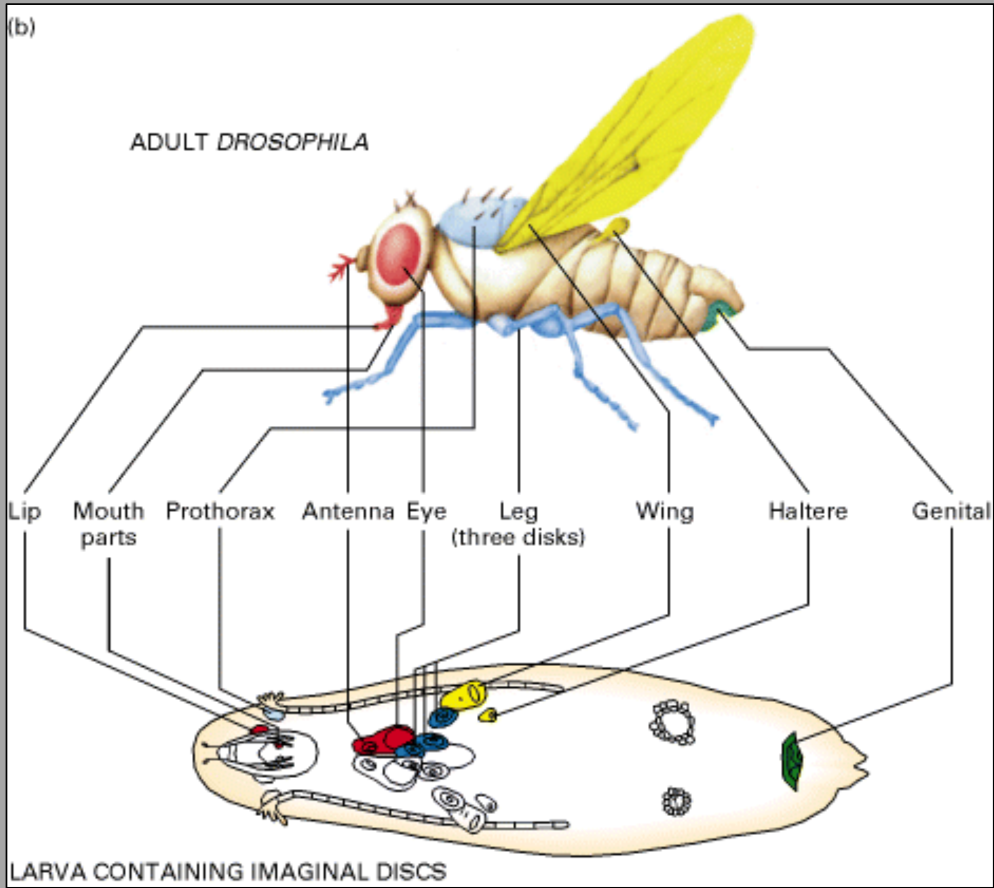
A metilação na linha germinal e no zigoto faz parte de um processo complexo de reprogramação epigenética que apaga a maioria das marcas parentais para garantir a totipotência, enquanto retém seletivamente outras. A metilação é estabelecida na linha germinal durante a gametogénese, enquanto o zigoto sofre desmetilação maciça, principalmente assimétrica, após a fertilização, seguida por metilação de novo após a implantação.

Ciclo de vida de *Drosophila*



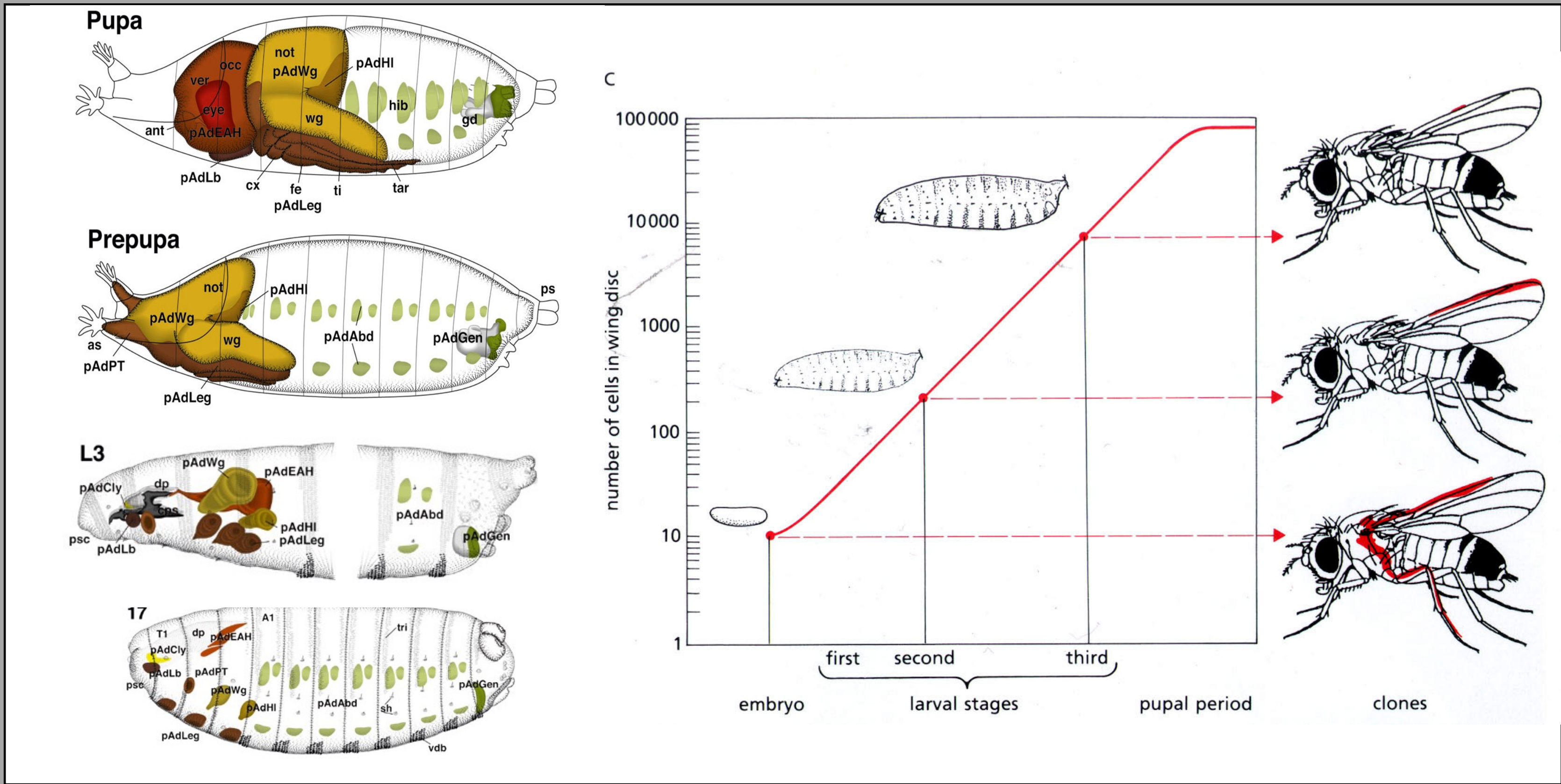
10 dias a 25ºC

18 dias a 18ºC

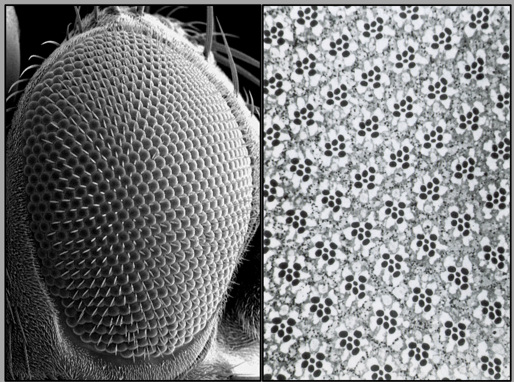


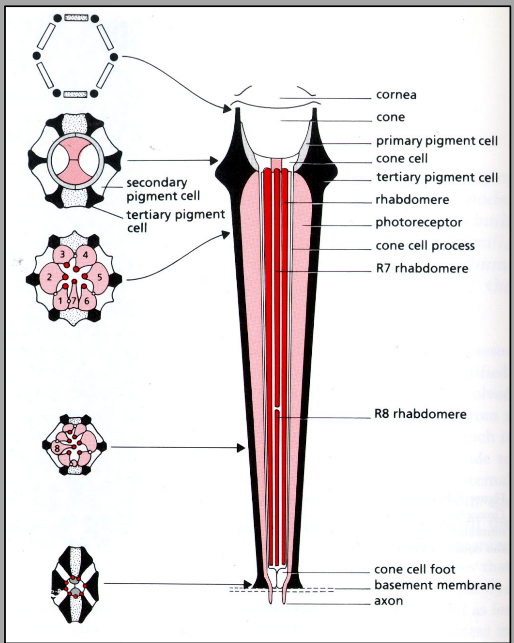
Durante a metamorfose dá-se uma profunda alteração do plano corporal e morfologia do animal que depende maioritariamente de grupos de células postos de lado durante a embriogénese e que crescem e se padronizam durante o período larvar. Na metamorfose serão poupados à destruição quase total dos tecidos larvares (CNS é parcialmente uma outra excepção) e, utilizando a energia proveniente das reservas larvares e da digestão dos tecidos crescem e formam o adulto.

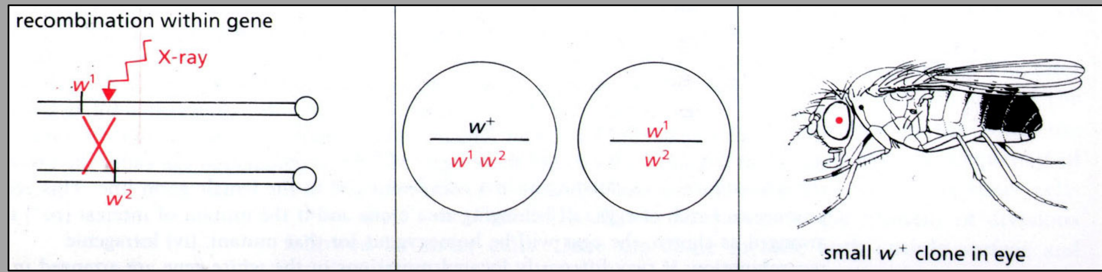
Parâmetros de crescimento podem ser determinados através de linhagem

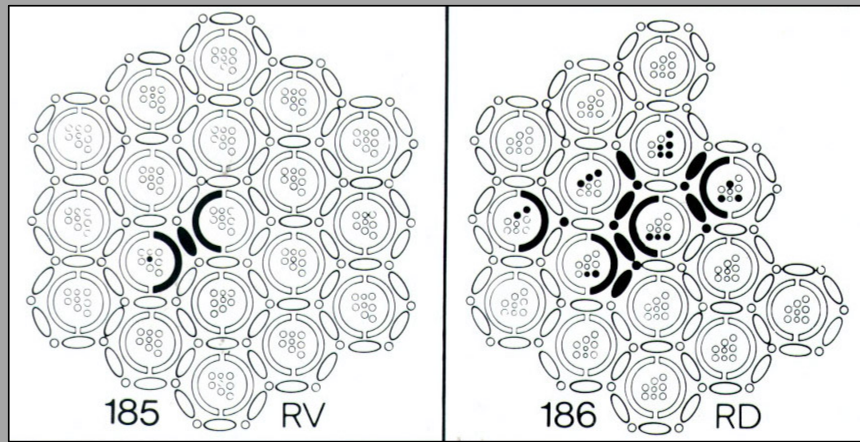


O olho composto da mosca é composto por unidades repetitivas chamadas omatídeos





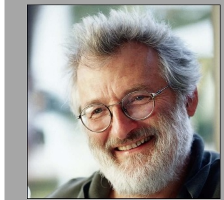


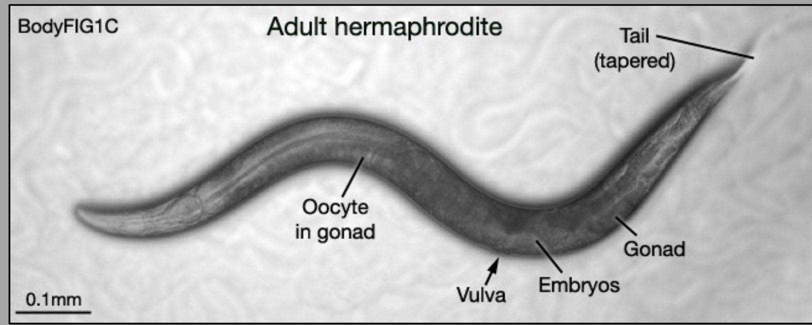


O olho composto de *Drosophila*
é composto de 800 omatídeos, cada um contendo 8
fotoreceptores, 4 células cone e células pigmentadas.

Para estabelecer linhagens é fundamental que compreendamos a morfogénese do organismo

John Sulston





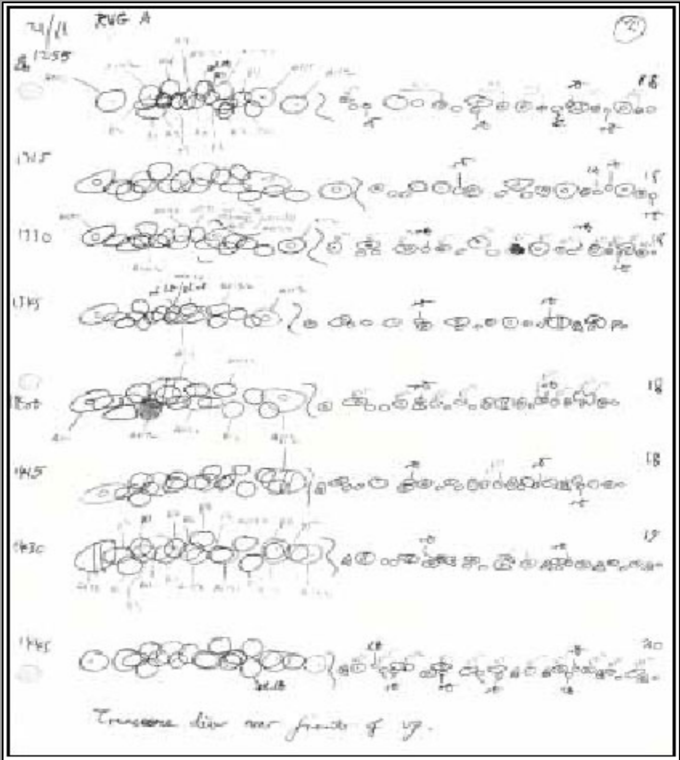
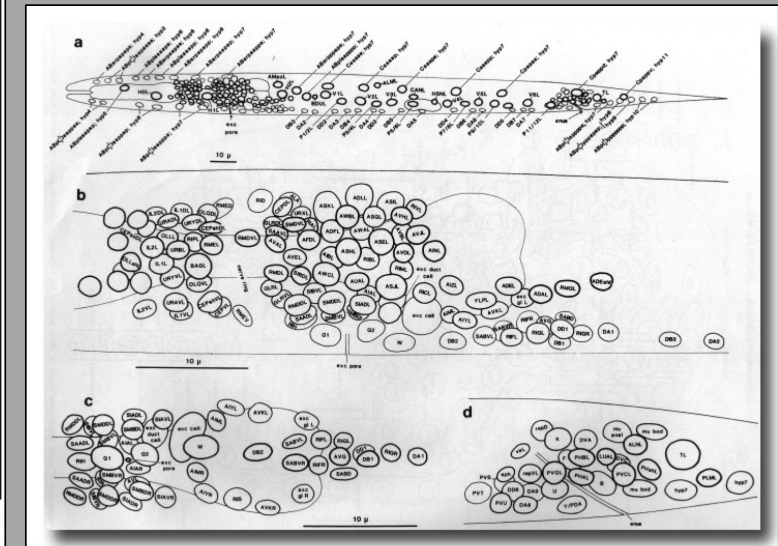
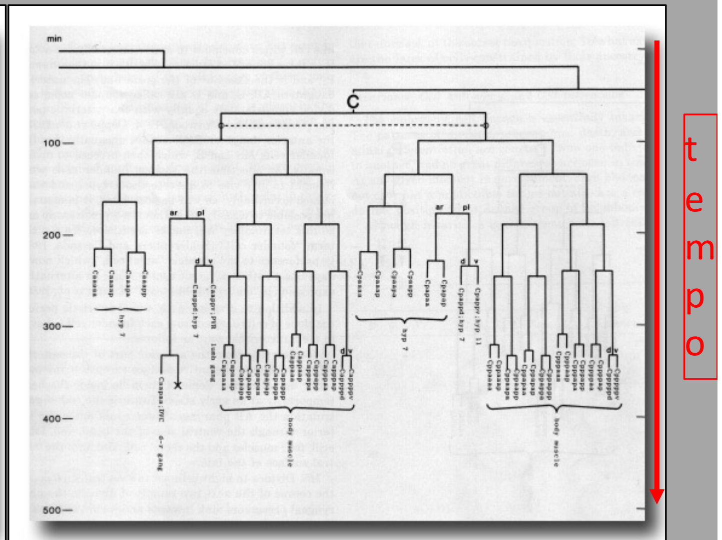
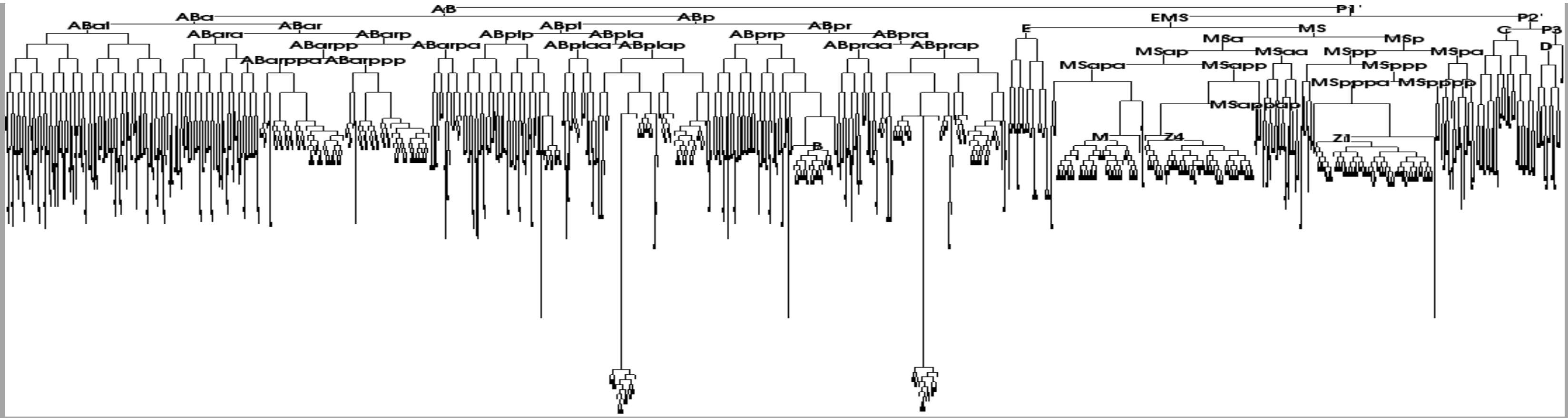


Figure 5. Drawings recording neuroblast divisions over a two hour period in the anterior ventral cord and retrovesicular ganglion, 1974.

Em *C. elegans*, a linhagem é invariante



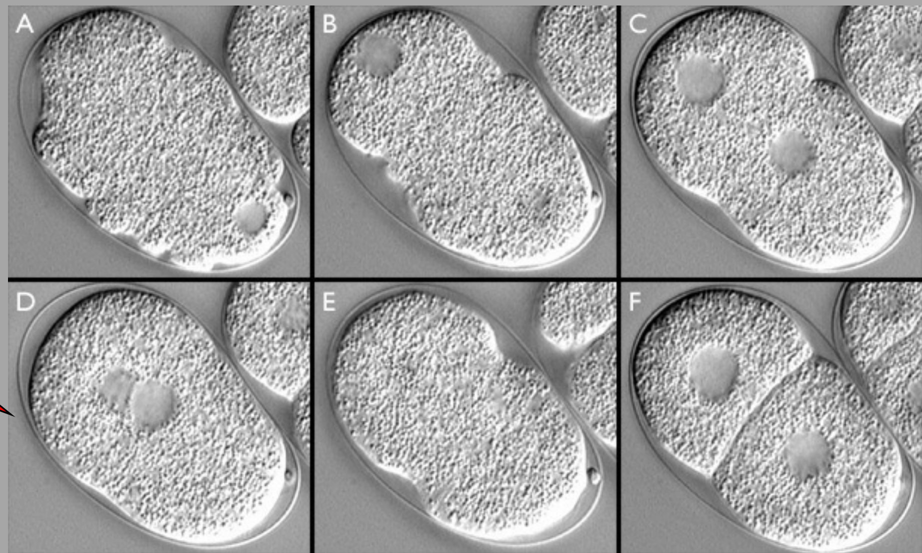


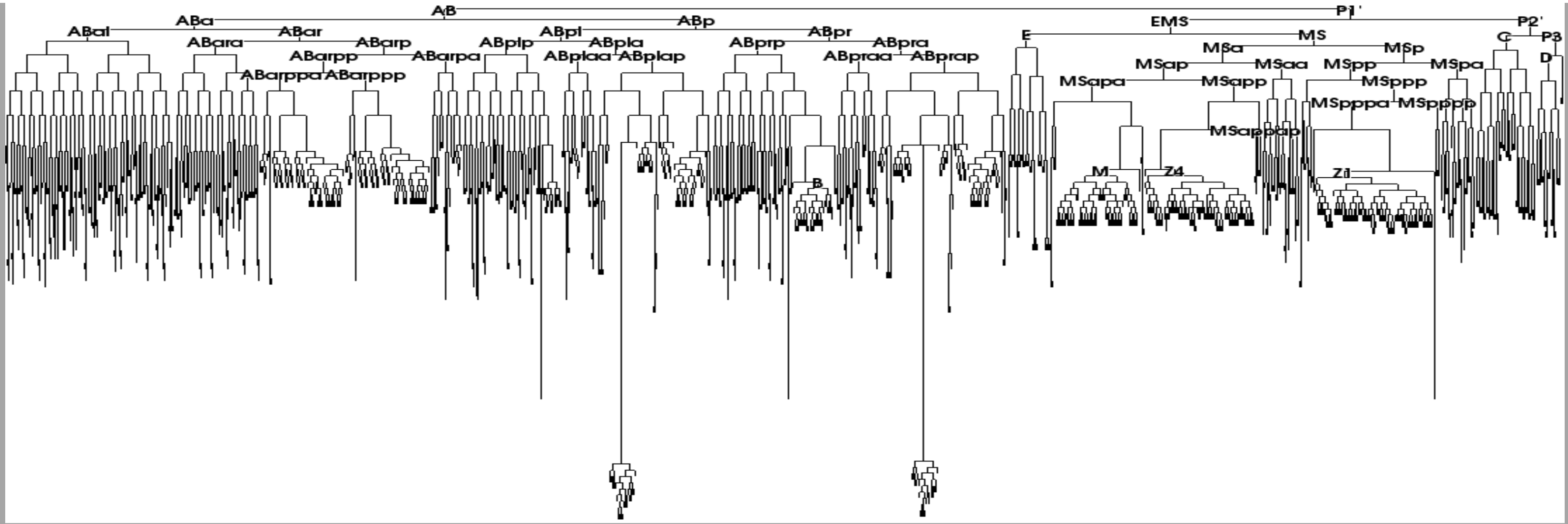


Embrião transparente

Linhagem invariável qu
leva a cerca de 959, 1031
células somáticas no adulto
hermafrodita e macho,
respectivamente

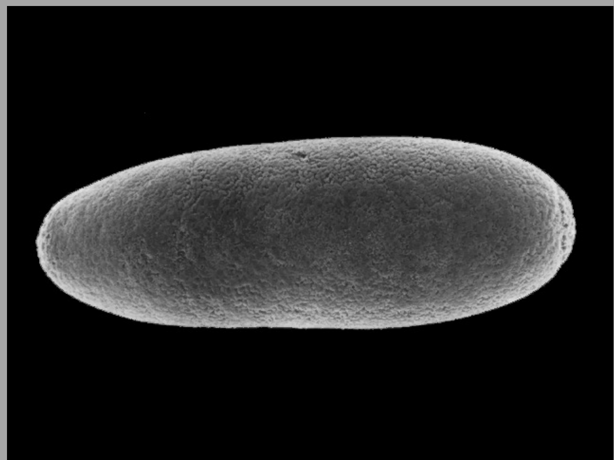
Laser



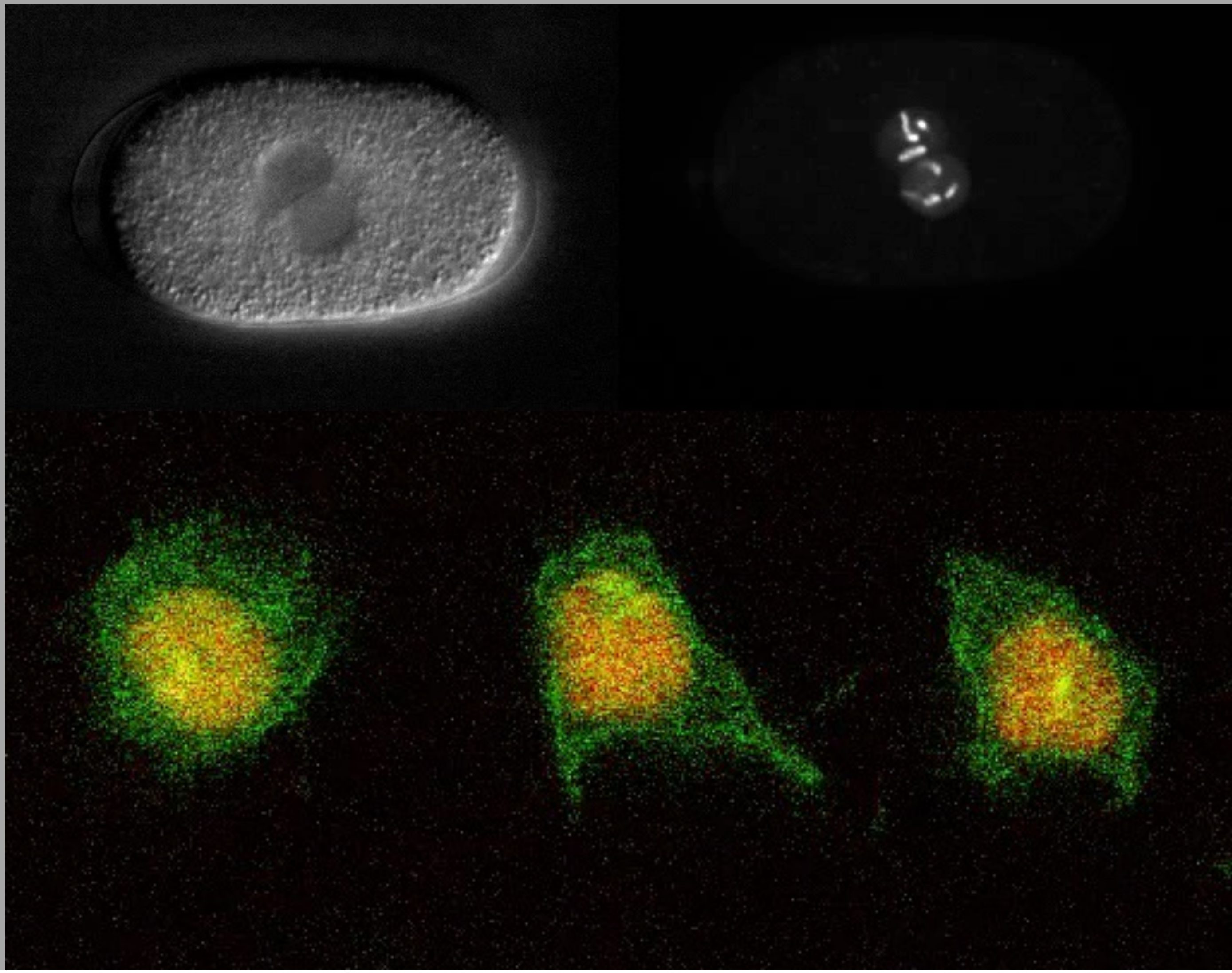


C. elegans

D. melanogaster



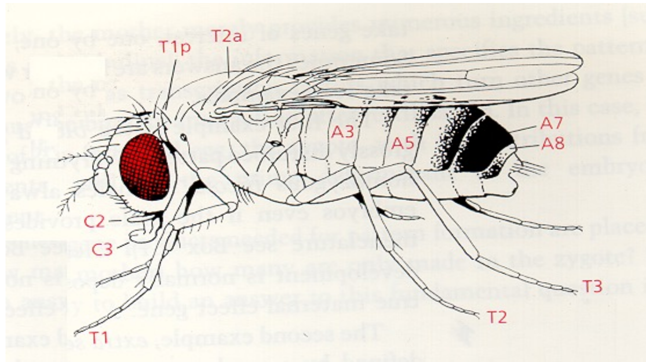
Distribuição assimétrica de determinantes (proteínas e mRNAs)
é fundamental para o estabelecimento dos destino celulares



Embrião de *C. elegans*

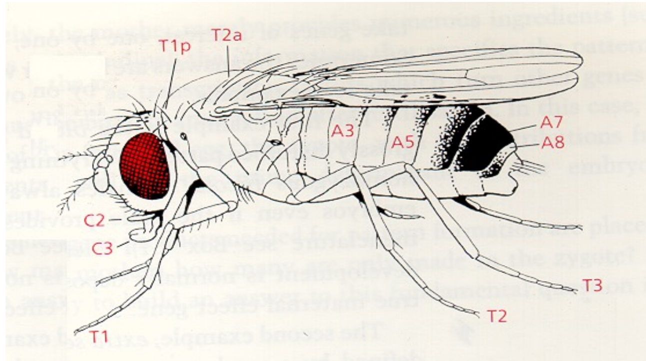
Divisão de neuroblasto de *Drosophila*

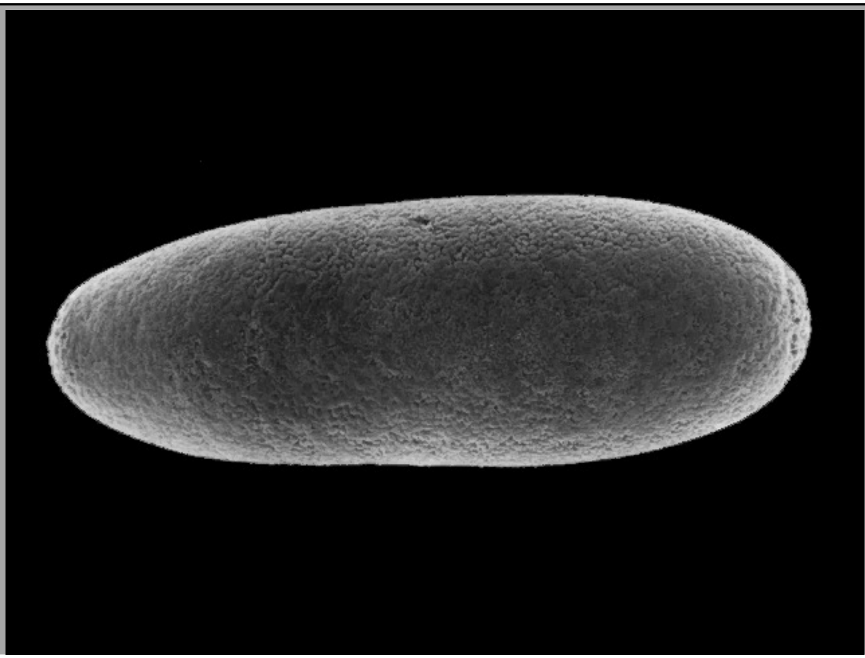
Zigoto → **24 h** → **Larva** → **9 dias** → **Adulto**

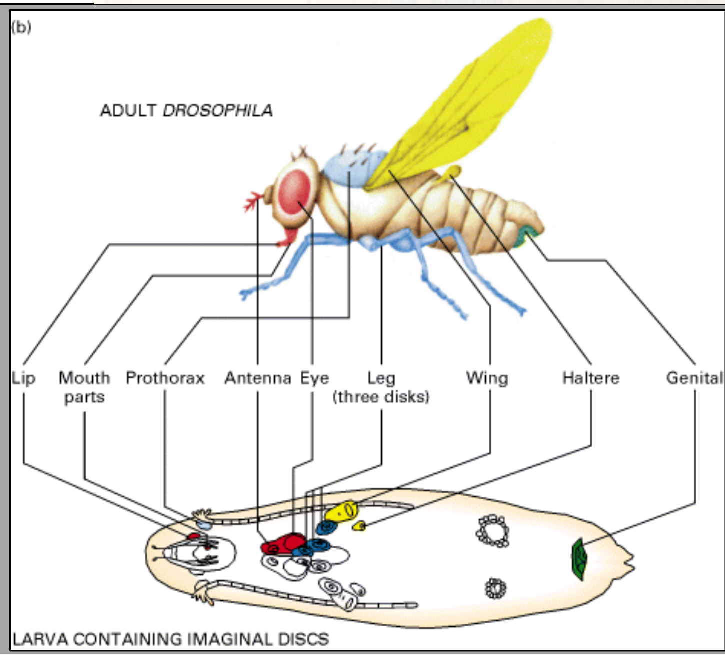


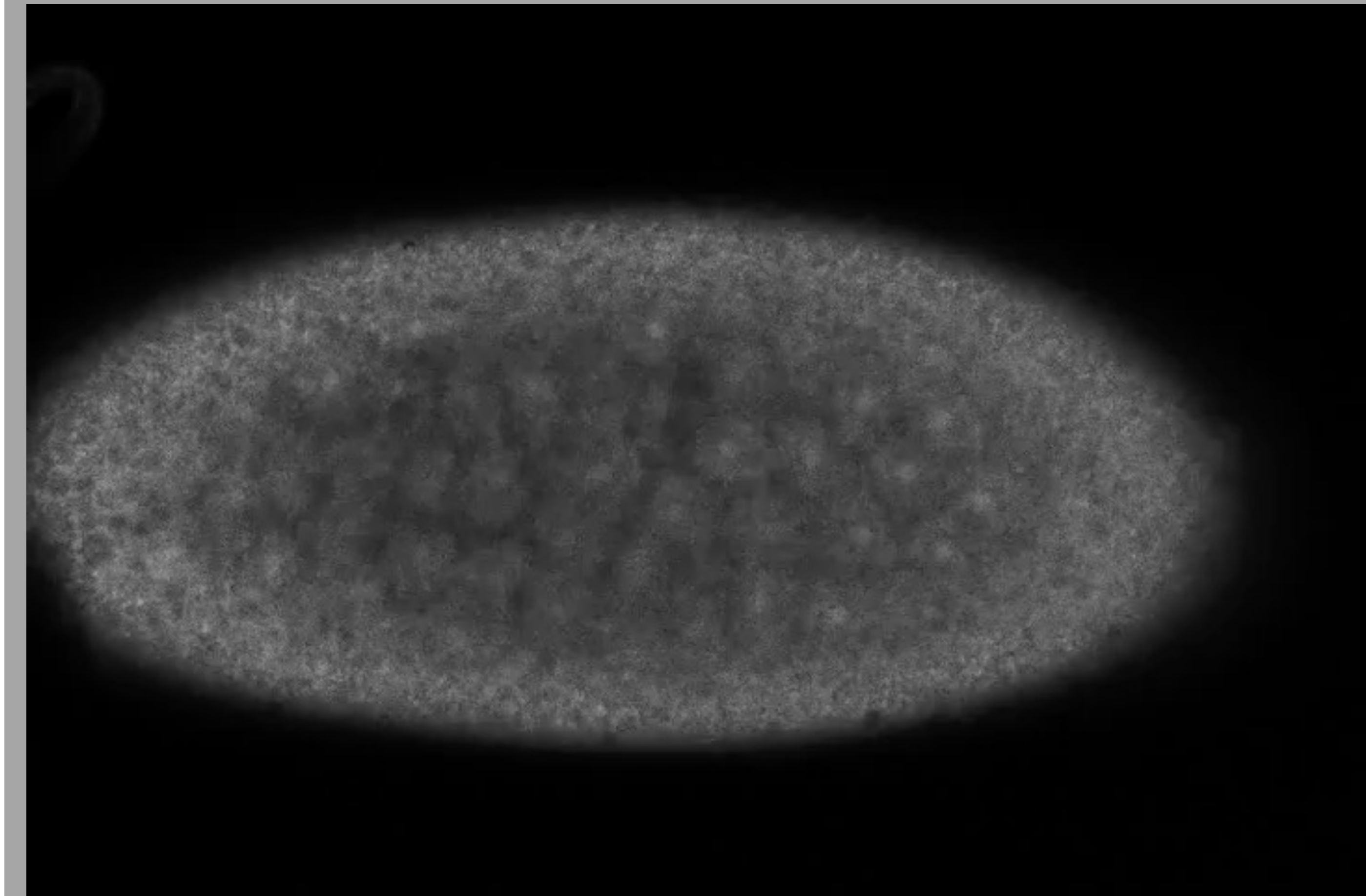
Zigoto → **Larva** → **Adulto**

24 h **9 dias**







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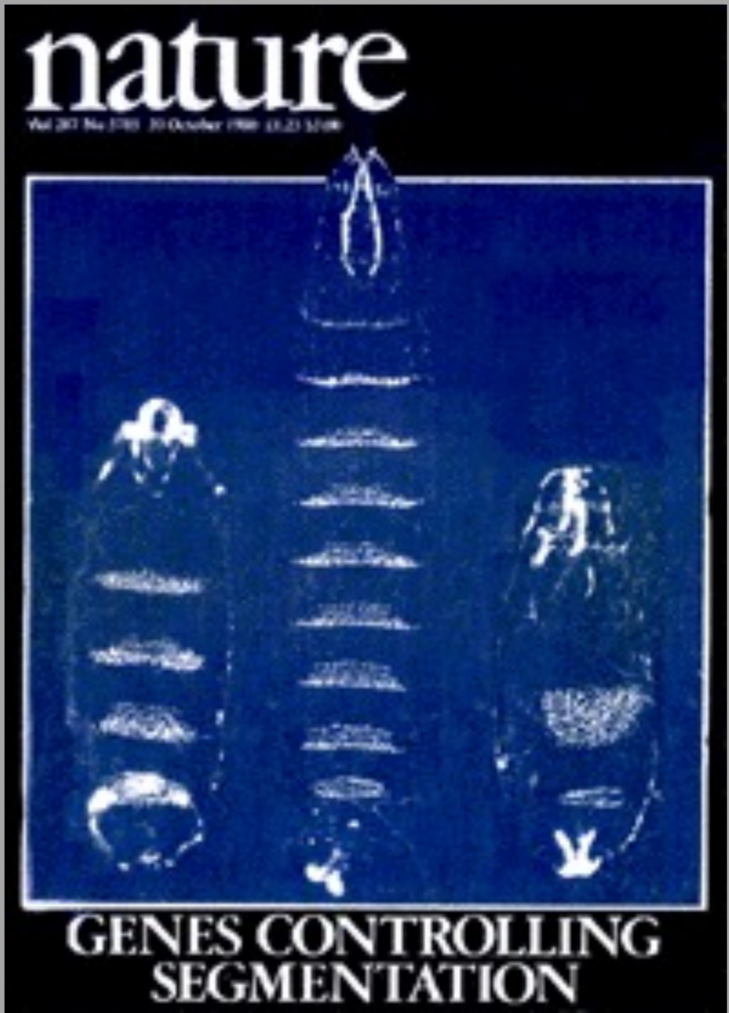
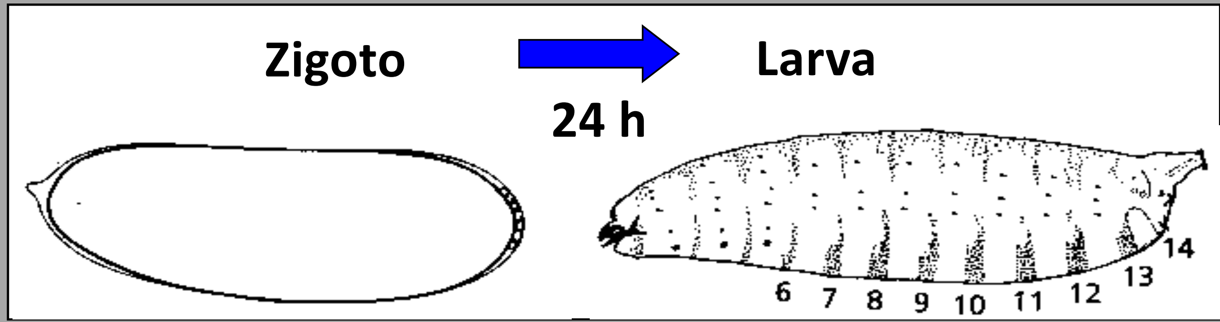


(1980)



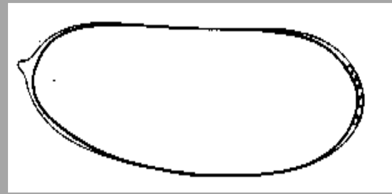
Prémio Nobel 1995

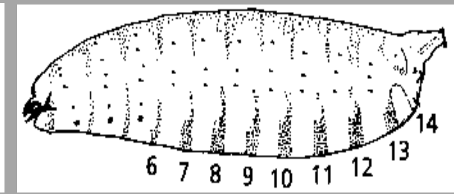
Ed Lewis, Christiane Nüsslein-Volhard & Eric Wieschaus

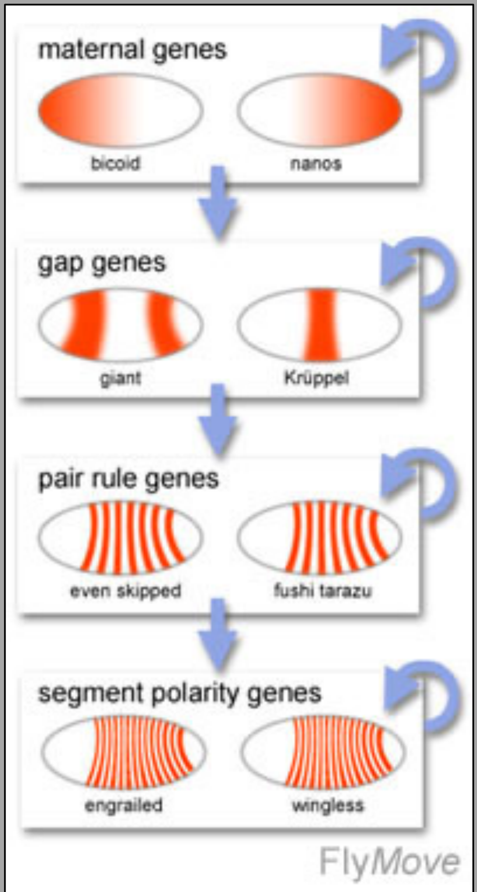
Hierarquia de segmentação

Zigoto → Larva

24 h





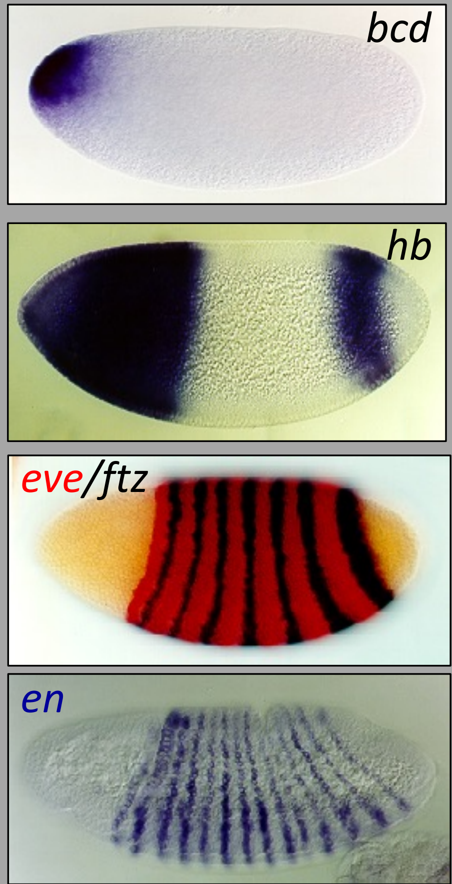


Genes maternos

Genes Gap

Genes controlo par

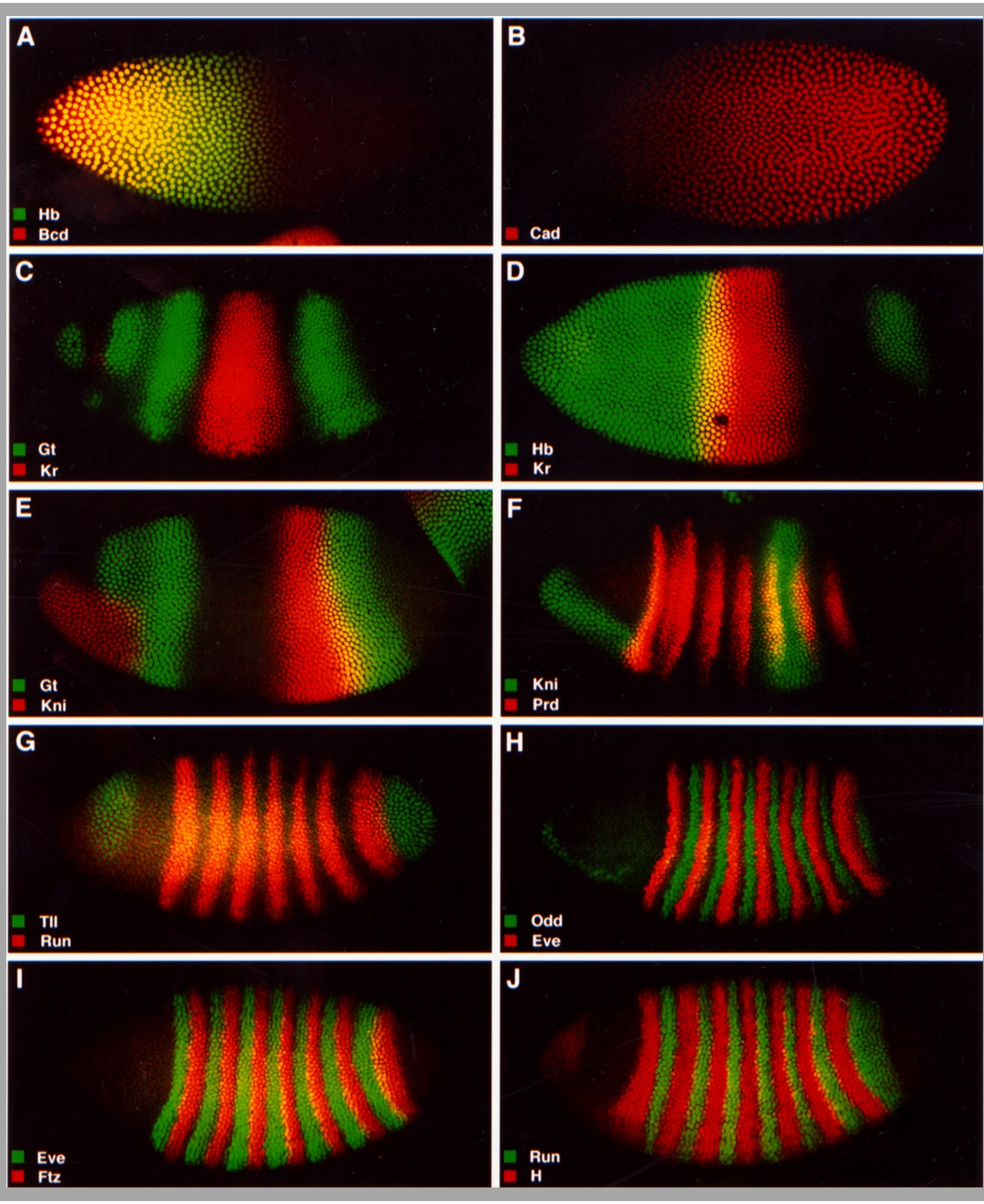
Genes de polaridade segmentar



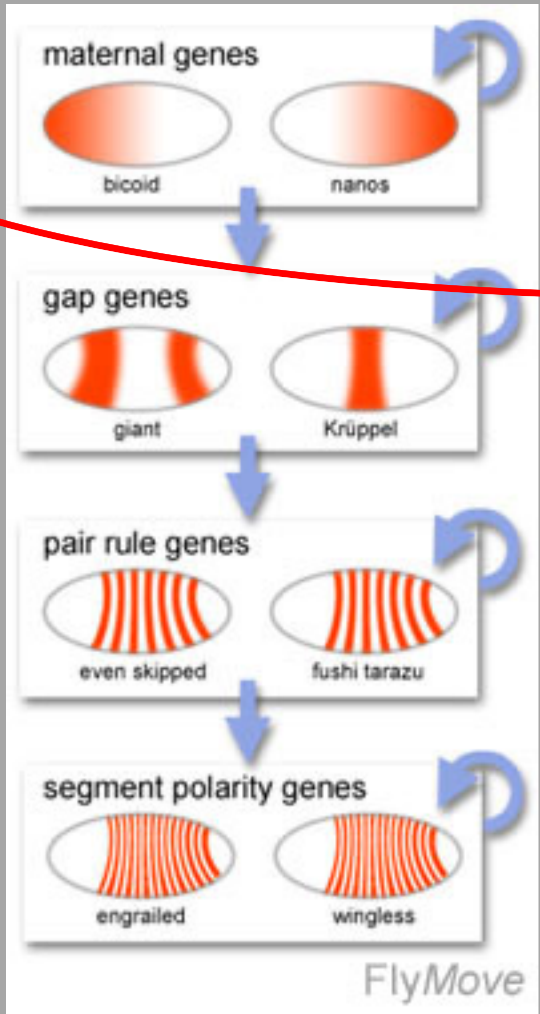
GENES MATERNOS

GENES GAP

GENES CONTROLO PAR



Hierarquia de segmentação



Genes maternos

bcd

Genes Gap

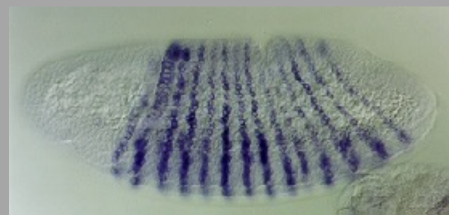
hb

Genes controlo par

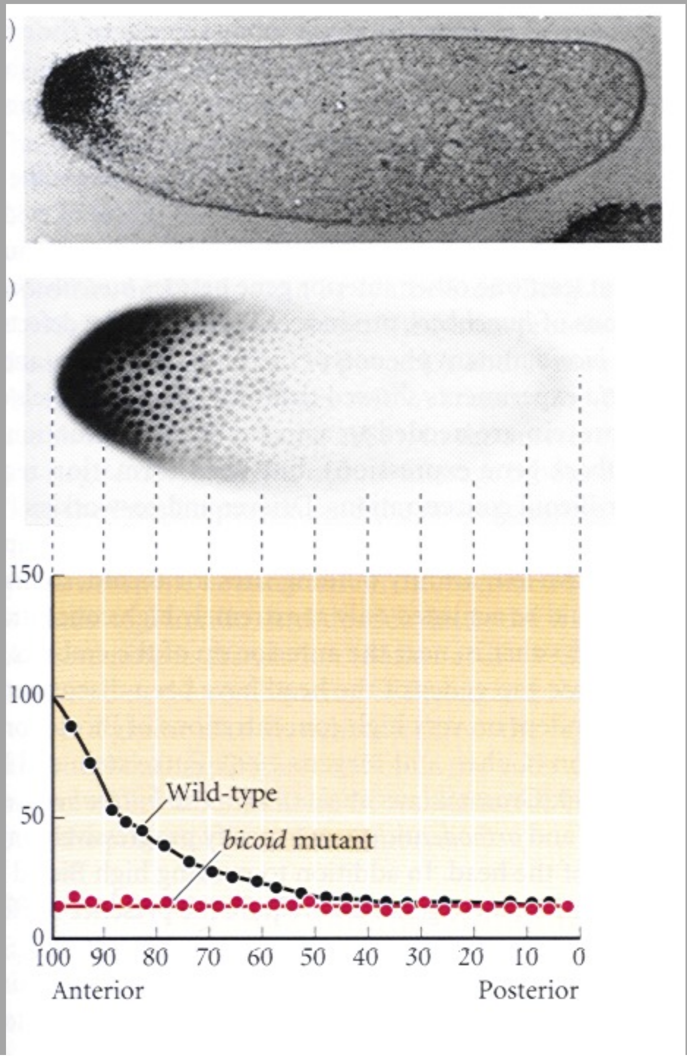
eve/ftz

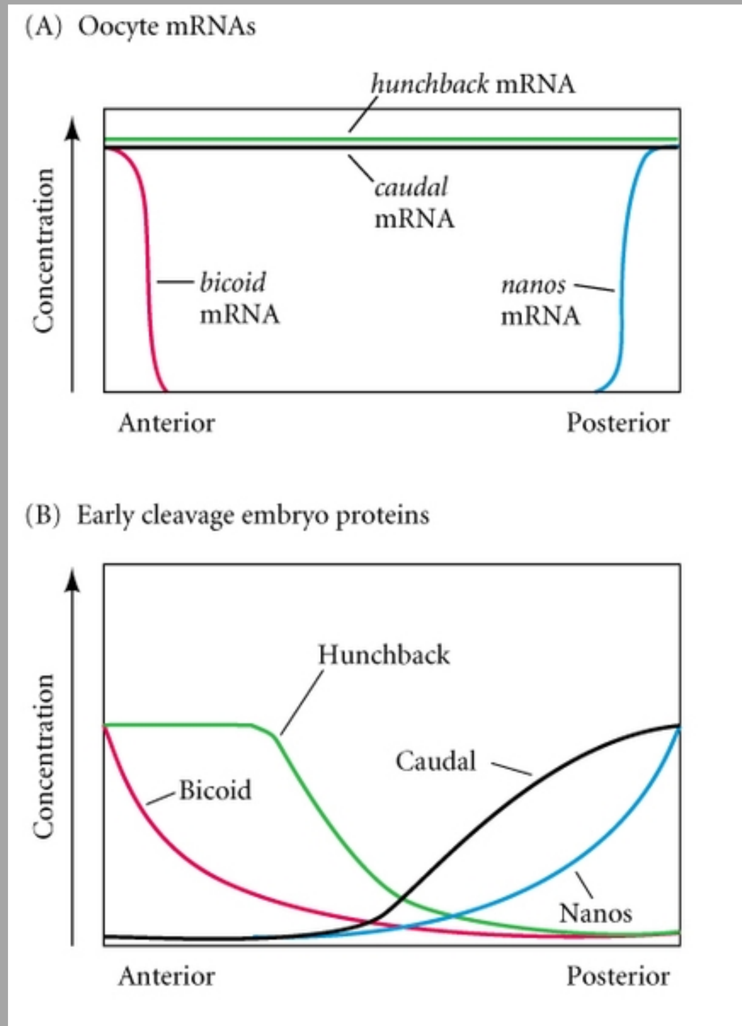
Genes de polaridade segmentar

en

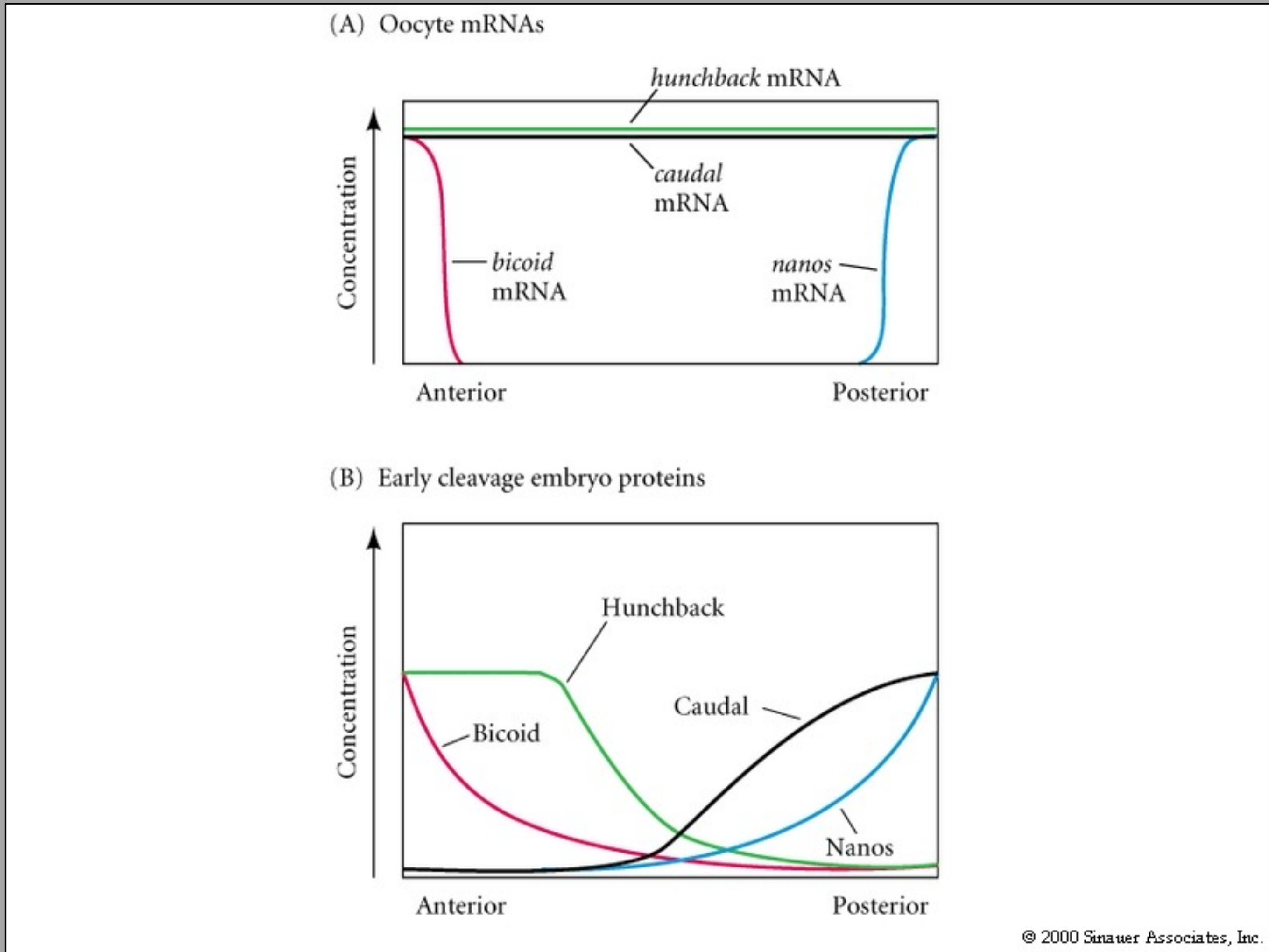


Os genes de efeito materno determinam dois “centros de organização” do embrião em posterior e anterior

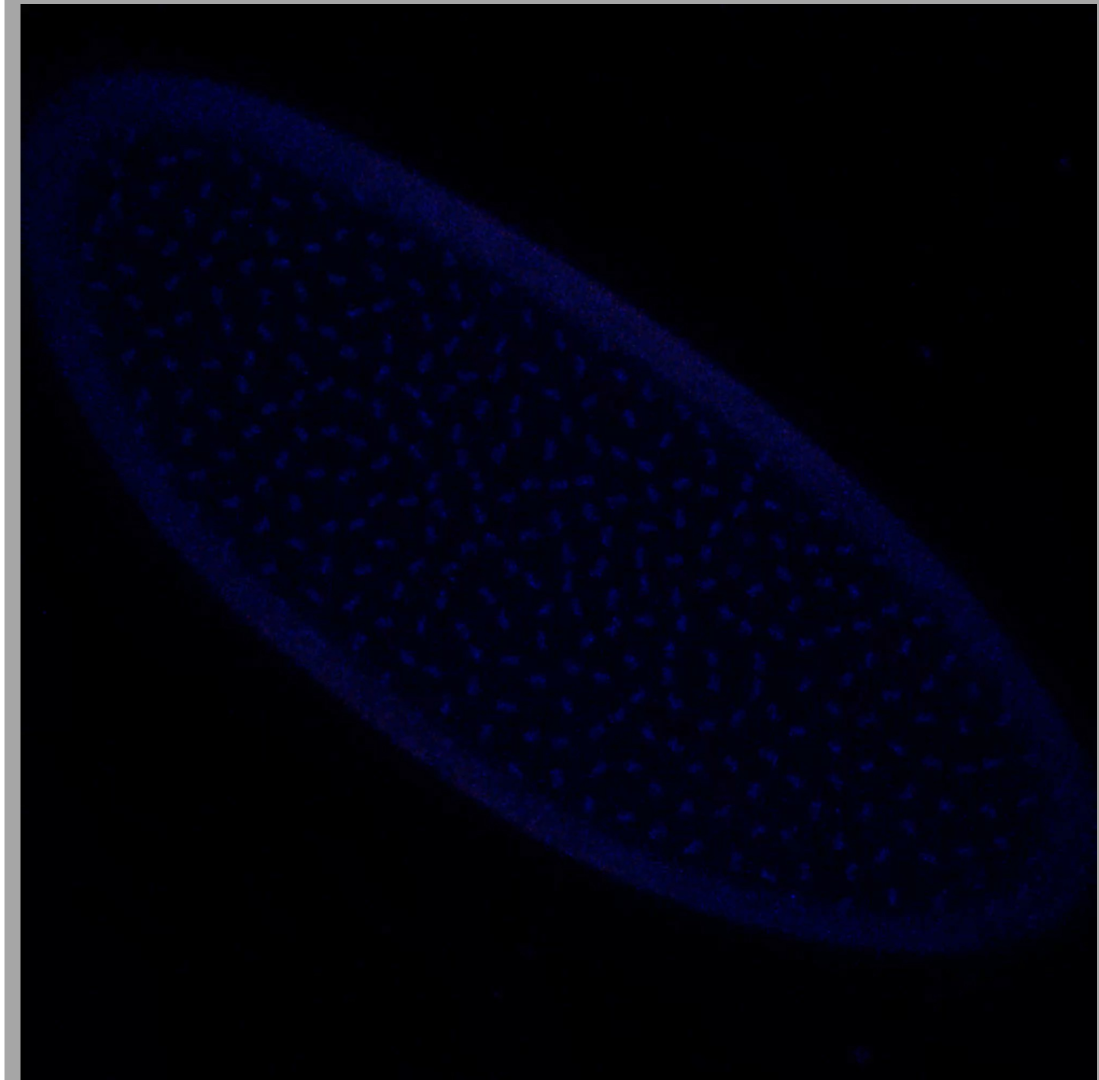


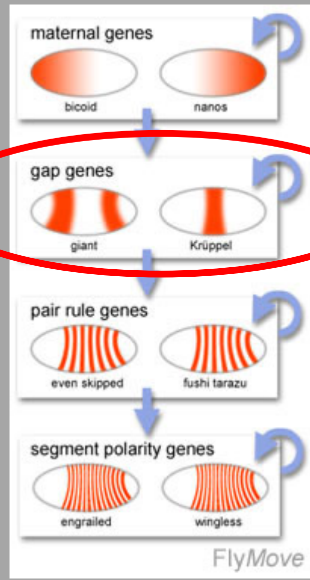


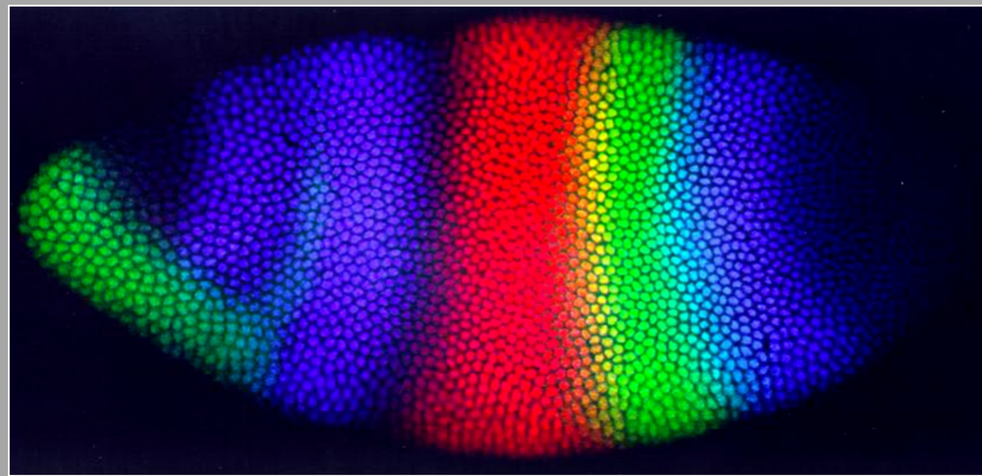
Os dois “centros de organização” do embrião de *Drosophila* (eixo A-P)



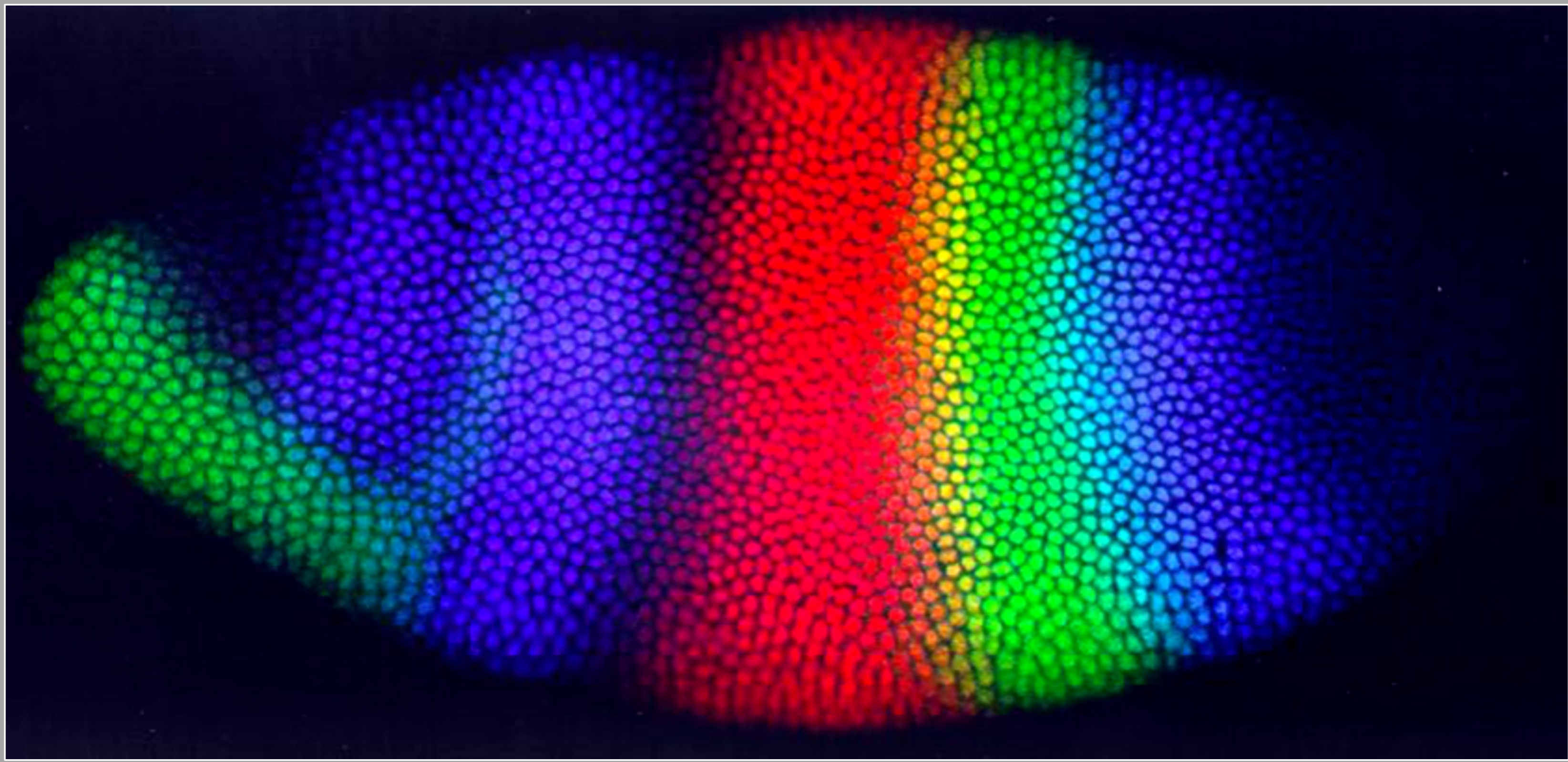
GENES GAP

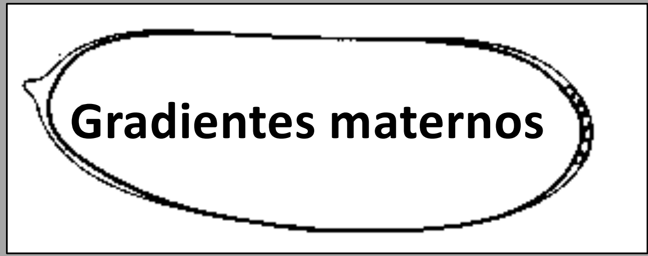



maternal genes
bicoid
nanos
gap genes
giant
Krüppel
pair rule genes
even skipped
fushi tarazu
segment polarity genes
engrailed
wingless
FlyMove



GENES GAP

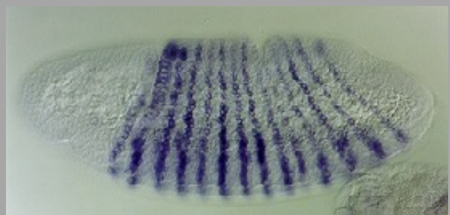




Gap genes

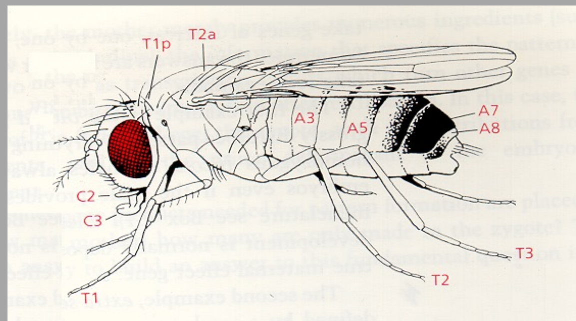
Segmentação

Pair-rule genes
Segment polarity genes

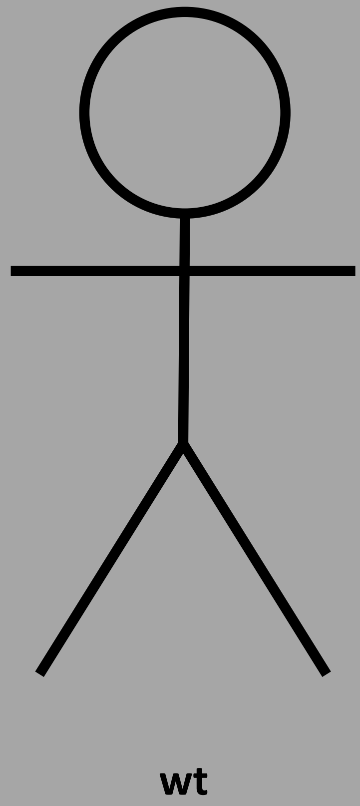


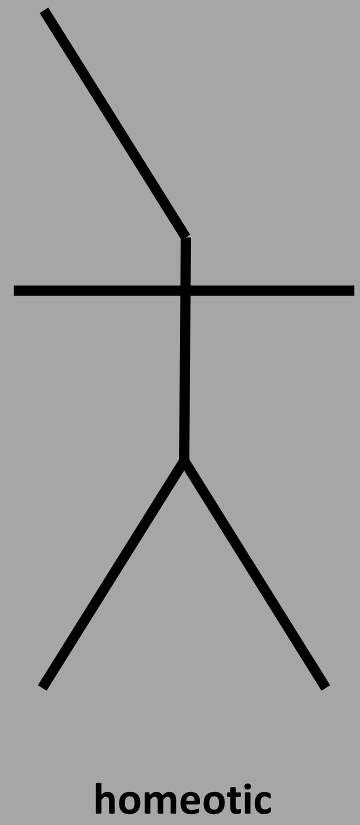
Diversidade segmentar

Hox genes

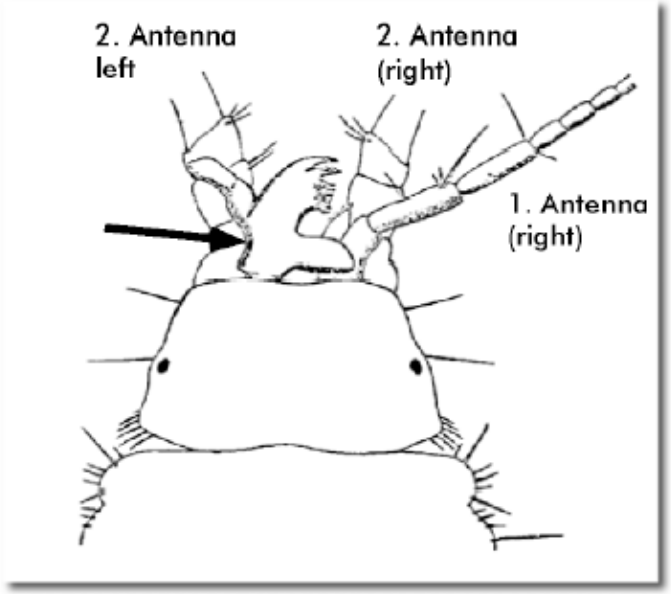


Homeotic mutations:





Mutações Homeóticas



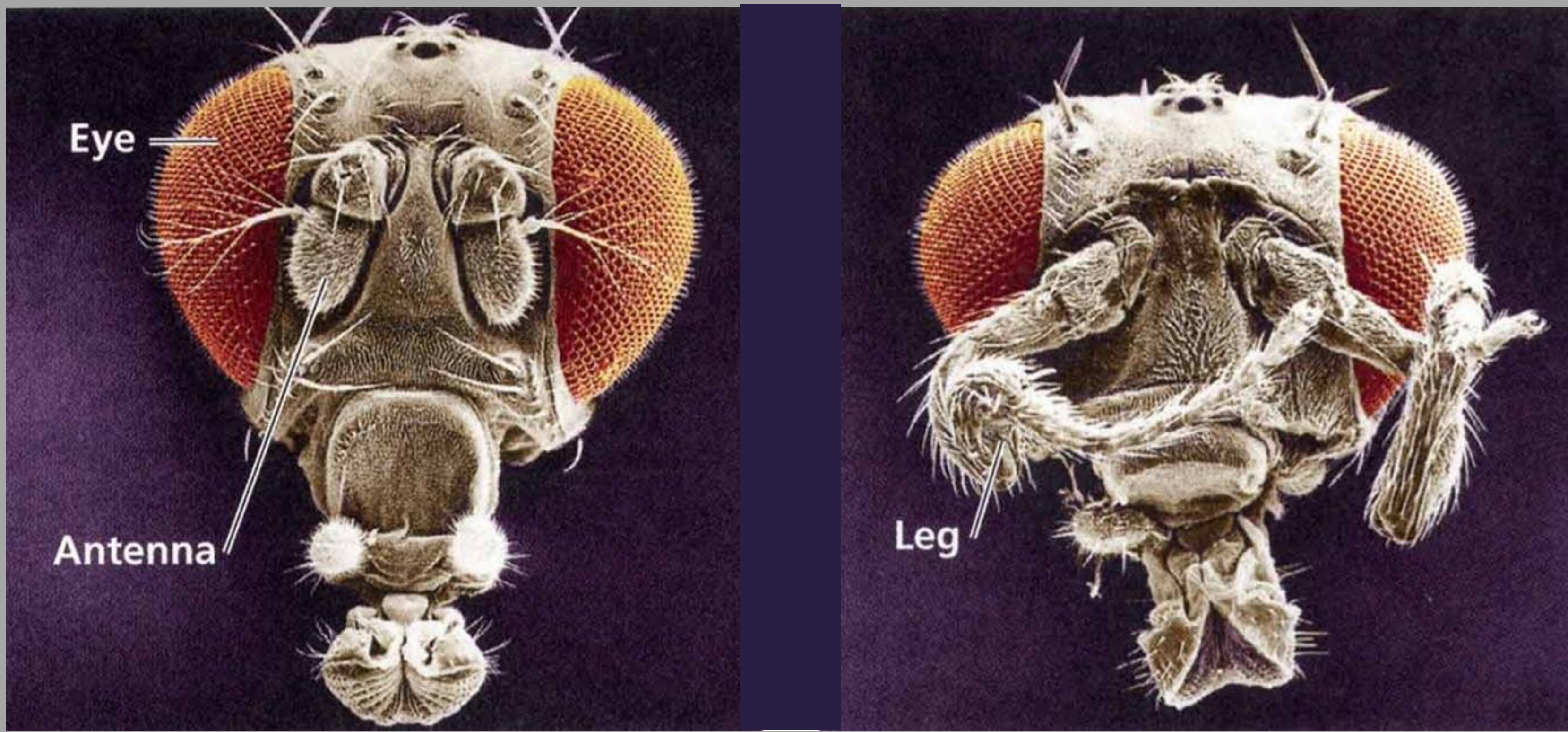
A Case of Homoeosis in a Crustacean of the Genus *Asellus*

[Proceedings of the Zoological Society, 1900]

BATESON DEFINES HOMEOSIS (1894)

"AN ORGAN TAKES ON THE LIKENESS OF ANOTHER ORGAN WITH WHICH IT IS IN SERIAL HOMOLOGY"

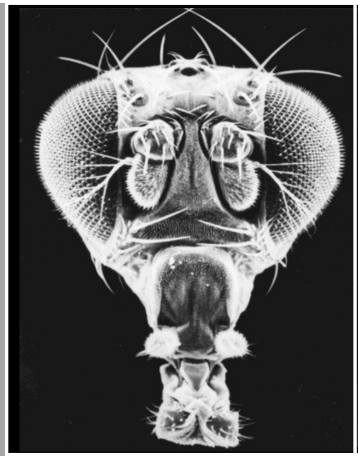
Homoeosis: A normal structure
at the wrong location



Genes homeóticos

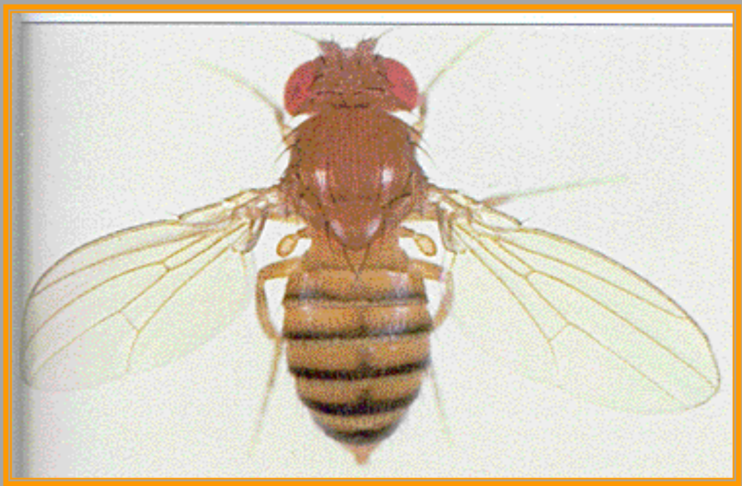
(Edward Lewis)

Genes que, quando mutados, levam a transformação de uma parte do corpo noutra, ou mais precisamente de um segmento em outro.



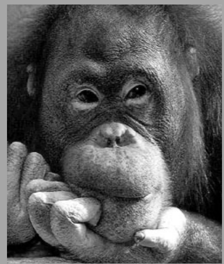
Mosca normal

Mutação *Antennapedia*



Mosca normal

Mutação *Ultrabitorax*

Phenotype = Genetics + Environment + GxE

$$(V_p = V_g + V_e + V_{gxe})$$